

This is the ultimate, zero-RAM-overhead security architecture. By offloading the heavy AI video processing to the Tapo C210's internal chip, your Raspberry Pi is completely protected from thermal throttling and memory crashes.

Here is your exact A-to-Z deployment plan to build this redundant system.

## **Phase 1: Storage & Initial Setup**

For 24/7 continuous recording, standard microSD cards will burn out in a few months. You must purchase a **High Endurance microSD card** (specifically branded as "High Endurance" by SanDisk or Samsung, typically Class 10 / U3).

1. Insert the 256GB High Endurance SD card into the Tapo C210.
2. Plug in the camera and open the **Tapo app** on your smartphone to connect it to your local Wi-Fi.
3. In the Tapo app, go to **Storage & Recording**, format the SD card, and set the recording schedule to **Continuous (24/7)**.
4. **Unlock the RTSP Stream:** In the Tapo app, go to **Advanced Settings > Camera Account**. Create a custom username and password (do not use your main Tapo account password). This action permanently unlocks the local video stream for your Raspberry Pi.

## **Phase 2: Internal AI Tracking & Google Voice**

We are going to let the camera do all the mechanical work.

1. In the Tapo app, go to **Detection & Alerts**. Turn on **Motion Detection** and **Person Detection**.
2. Turn on **Pan/Tilt Tracking**. The camera will now automatically rotate its motors to follow anyone walking through your front door without your Pi server having to calculate the movement.
3. **Google Home Integration:** Open your Google Home app, click the + button, select **Works with Google**, and search for **Tapo**. Log in to link the accounts. You can now say, "Hey Google, show me the front door camera" to stream the feed to any smart display or Chromecast.

## **Phase 3: Home Assistant Integration**

Since the 2024.2 update, Home Assistant has a native TP-Link integration, meaning no custom HACS hacking is required.

1. Open Home Assistant, go to **Settings > Devices & Services > Add Integration**.
2. Search for **TP-Link Smart Home** (or Tapo).
3. Home Assistant will auto-discover the C210 on your network. Log in with the credentials you made in the app.
4. Your camera's PTZ controls, privacy mode, and motion sensors will now appear as native buttons and switches in your dashboard.

## **Phase 4: The Raspberry Pi SSD Automation (Zero-RAM)**

(Note: Do not execute this step until you have successfully replaced your failing CCTV SSD and

*correctly re-mounted it to /mnt/cctv\_ssd, or the Pi will crash trying to write to a dead drive!)*

Instead of running a heavy AI Docker container on the Pi, you will use the official integration's binary sensor to trigger a lightweight recording script.

In Home Assistant, create a new Automation with this exact logic:

- **Trigger:** When binary\_sensor.front\_door\_motion changes from Off to On.
- **Action:** Call Service: shell\_command.record\_cctv\_motion

In your Pi's configuration.yaml, define that shell command to simply rip the RTSP stream directly to your external drive for 60 seconds using FFmpeg:

```
YAML
shell_command:
  record_cctv_motion: "ffmpeg -i
rtsp://YOUR_CAMERA_ACCOUNT:PASSWORD@IP_ADDRESS:554/stream1 -c copy -t 60
/mnt/cctv_ssd/front_door_$(date +%Y%m%d_%H%M%S).mp4"
```

*(Because -c copy is used, the Pi does zero video encoding. It just writes the raw data to the SSD, using virtually 0% CPU).*



## Phase 5: The Dashboard Intercom Workaround

Because RTSP does not support sending audio backward into the camera, you will use the deep-link trick for your intercom.

Add a new **Button Card** to your Home Assistant dashboard:

- **Name:** Front Door Intercom
- **Icon:** mdi:microphone
- **Tap Action:** Navigate to URL
- **URL:** app://com.tplink.iot

When you see someone at the door on the HA dashboard, tap that shortcut on your Samsung device. The phone will instantly minimize Home Assistant, launch the Tapo app, and allow you to hold the microphone button to speak.

To finalize the Phase 4 recording setup, are you currently using MotionEye, Frigate, or just raw FFmpeg shell commands to manage your existing video files on that Pi server?